Artificial Intelligence

Machine Learning & Deep Learning

Week 1 - Day 1

Introduction to AI & Python Fundamentals

Learning Objectives

1. Understand the fundamentals of Artificial Intelligence and its applications
2. Learn the difference between AI, Machine Learning, and Deep Learning
3. Explore real-world AI applications across various industries
4. Set up Python development environment (Anaconda, VSCode, PyCharm)
5. Master Python basic syntax, variables, and data types
6. Write your first Python programs and execute them successfully

What is Artificial Intelligence?

Artificial Intelligence (AI)

The simulation of human intelligence processes by machines, especially computer systems. These processes include learning, reasoning, problem-solving, perception, and language understanding.

Key Characteristics of AI:

Learning

Ability to acquire and apply knowledge

Reasoning

Drawing inferences and making decisions

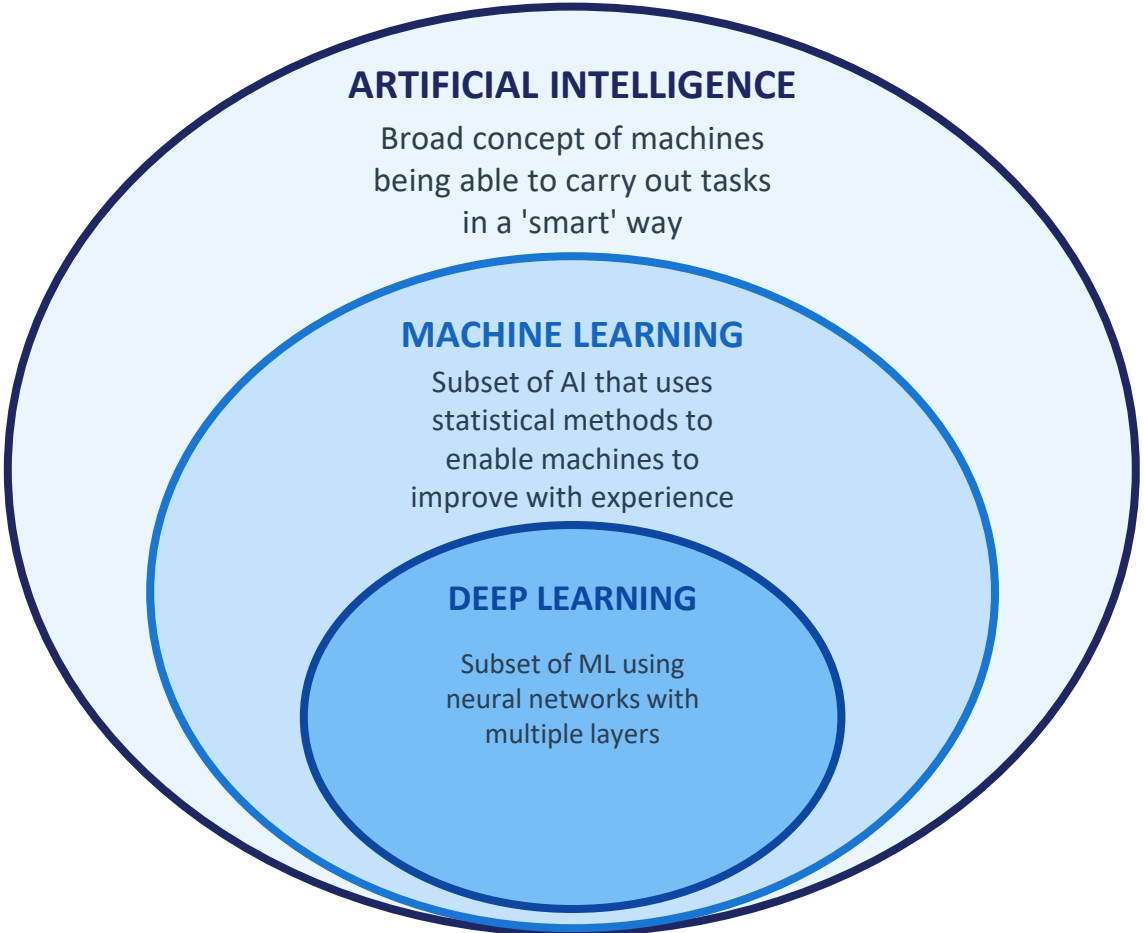
Problem Solving

Finding solutions to complex challenges

Perception

Interpreting sensory data (vision, speech)

AI vs Machine Learning vs Deep Learning



Types of Artificial Intelligence

Based on Capabilities:

Narrow AI (Weak AI)

Designed for specific tasks
Examples: Siri, Alexa, Chess AI

General AI (Strong AI)

Human-like intelligence
Can understand and learn any task

Super AI

Surpasses human intelligence
Theoretical concept

Based on Functionality:

Reactive Machines

No memory, purely reactive
Example: Deep Blue (chess)

Limited Memory

Uses past experience
Example: Self-driving cars

Theory of Mind

Understands emotions (future)
Social interaction capabilities

Self-Aware AI

Conscious machines (future)
Self-awareness and emotions

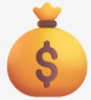
Real-World AI Applications

Healthcare

Disease diagnosis, Drug discovery, Medical imaging

Transportation

Self-driving cars, Traffic prediction, Route optimization



Finance

Fraud detection, Trading algorithms, Credit scoring

E-Commerce

Product recommendations, Chatbots, Demand forecasting

Entertainment

Netflix recommendations, Game AI, Content creation

Manufacturing

Quality control, Predictive maintenance, Robotics

AI Career Opportunities in Pakistan

Companies Hiring AI/ML Talent:

Careem	Afiniti	Addo.ai	Arbisoft
I2c	Xavor	Fiverivers	Confiz
Crossover	NetSol	Research Institutes	Tech Startups

Career Paths After This Course:

- ✓ AI Associate Engineer
- ✓ Machine Learning Associate Analyst
- ✓ Assistant Data Analyst
- ✓ Research Assistant

Average Starting Salary: PKR 50,000 - 100,000/month

Course Structure & Overview

Duration: 3 Months | 4 Hours/Day | Theory: 20% | Practical: 80%

Week 1-2	Python Fundamentals & OOP
Week 3-4	Statistics, NumPy, Pandas, Visualization
Week 5-6	Machine Learning & NLP Basics
Week 7-9	Deep Learning (CNN, RNN, LSTM)
Week 10-12	Projects & Microsoft Azure AI

Tools & Technologies You'll Master:

Python • NumPy • Pandas • Matplotlib • Scikit-learn • TensorFlow • PyTorch • Keras • Azure AI

Setting Up Your Development Environment

Anaconda

Python distribution with 250+ packages — *anaconda.com* — **Essential**

VSCode

Lightweight, powerful code editor — *code.visualstudio.com* — **Recommended**

PyCharm Community

Professional Python IDE — *jetbrains.com/pycharm* — **Optional**

PyTorch

Deep learning framework — *pytorch.org* — **Week 7+**

TensorFlow 2.0

Google's ML framework — *tensorflow.org* — **Week 7+**

Why Python for AI & Machine Learning?

Python - The Language of AI

Rich Libraries

NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch - everything you need

Easy to Learn

Simple syntax, readable code, great for beginners and experts

Large Community

Millions of developers, extensive documentation, active support

Rapid Development

Quick prototyping, testing, and deployment of AI models

Versatile

Web development, data analysis, automation, AI/ML - all in one

Industry Standard

Used by Google, Facebook, Netflix, NASA, and thousands more

Types

What is a Variable?

A variable is a container for storing data values. Think of it as a labeled box where you can put information and retrieve it later.

Variable Assignment Examples:

```
# Numbers
num1 = 1.5
num2 = 6.3
age = 25
```

```
# Strings
name = "Ahmed"
course = "AI/ML"
city = 'Lahore'
```

Basic Data Types in Python:

int	Whole numbers	*42, -10, 0*
float	Decimal numbers	*3.14, -0.5, 2.0*
str	Text/String	*"Hello", 'Python'*
bool	True/False	*True, False*

Python Operators

Arithmetic Operators

+	Addition	*5 + 3 = 8*
-	Subtraction	*10 - 4 = 6*
*	Multiplication	*3 * 4 = 12*
/	Division	*15 / 3 = 5.0*
**	Exponent	*2 ** 3 = 8*

Comparison Operators

==	Equal to	*5 == 5 → True*
!=	Not equal	*5 != 3 → True*
>	Greater than	*7 > 3 → True*
<	Less than	*2 < 5 → True*
>=	Greater/Equal	*5 >= 5 → True*

Practical Example:

```
# Calculate area of a rectangle
length = 10
width = 5
area = length * width
print(f"The area is {area}")  # Output: The area is 50
# Check if number is positive
num = 15
```

String Operations in Python

Strings are sequences of characters used to store text. You can use single ('...') or double quotes ("...").

Creating Strings:

```
name = "Ahmed Ali"
message = 'Welcome to AI Course'
multiline = """This is a
multi-line string"""
```

Common String Operations:

Concatenation

`first + " " + last → "Ahmed Ali"`

Length

`len("Python") → 6`

Replace

`text.replace("old", "new") → new text`

Repetition

`"AI" * 3 → "AIAIAI"`

Upper/Lower

`name.upper() → "AHMED"`

Practical Example:

```
greeting = "Hello"
name = "Student"
message = greeting + ", " + name + "!"
```

User Input & Type Casting

Getting User Input:

```
name = input("Enter your name: ")
print("Hello, " + name)
# Note: input() always returns a string!
```

Type Casting (Converting Data Types):

`int()`	Convert to integer	*int("42") → 42*
`float()`	Convert to float	*float("3.14") → 3.14*
`str()`	Convert to string	*str(100) → "100"*
`bool()`	Convert to boolean	*bool(1) → True*

Complete Example:

```
age_str = input("Enter your age: ")
age = int(age_str)  # Convert string to integer
print(f"Next year you will be {age + 1} years old")
```

Python Comments

Why Use Comments?

📝 Explain complex code logic

👥 Help other developers understand your code

🔍 Document functions and classes

🚫 Temporarily disable code during debugging

Types of Comments:

1. Single-line Comments (use #)

```
# This is a single-line comment
x = 10  # This is also a comment
# Comments are ignored by Python
```

2. Multi-line Comments (use """...""")

```
"""
This is a multi-line comment.
It can span multiple lines.
Useful for documentation.
"""
```

Best Practices:

✓ Keep comments short and relevant

✓ Use clear, descriptive variable names instead of excessive comments

✓ Update comments when you update code

✗ Don't state the obvious (avoid: x = 5 # assign 5 to x)

Your First Python Program

1. The Classic 'Hello World'

```
print("Hello, World!")
```

2. Program: Add Two Numbers

```
# This program adds two numbers
num1 = 1.5
num2 = 6.3

# Add two numbers
sum = num1 + num2

# Display the sum
print(f'The sum of {num1} and {num2} is {sum}')
```

Output:

```
The sum of 1.5 and 6.3 is 7.8
```

3. Program: Temperature Converter

```
celsius = 37.5
fahrenheit = (celsius * 1.8) + 32
print(f'{celsius}°C = {fahrenheit}°F')
# Output: 37.5°C = 99.5°F
```

Today's Practice Tasks

Complete These Tasks (from Course Annexure-I):

Task 1

Software Installation

Download and install Anaconda, VSCode, PyCharm

⏱ 30 mins

Task 3

Addition Program

Write a program that adds two numbers

⏱ 10 mins

Task 4

User Input

Get input from user and add two numbers

⏱ 15 mins

Task 5

Square Root

Calculate and display square root of a number

⏱ 15 mins

Task 7

Temperature Converter

Convert Celsius to Fahrenheit

⏱ 15 mins

Python in Real-World AI Applications

Netflix Recommendation Engine

Python + Machine Learning

Analyzes viewing patterns to suggest shows you'll love

Tesla Autopilot

Python + Deep Learning + Computer Vision

Self-driving technology using neural networks

Google Translate

Python + NLP + Neural Networks

Translates 100+ languages using AI

Spotify Discover Weekly

Python + Collaborative Filtering

Personalized music recommendations

Amazon Alexa

Python + Speech Recognition + NLP

Voice assistant understanding natural language

Instagram Filters

Python + Image Processing + CNN

Real-time face detection and AR effects

Day 1 Summary - What We Learned

Understanding AI

- Definition and importance of Artificial Intelligence
- Difference between AI, Machine Learning, and Deep Learning
- Types of AI: Narrow, General, and Super AI

Career Opportunities

- Job market in Pakistan (Careem, Afiniti, etc.)
- Career paths: AI Engineer, ML Analyst, Data Analyst
- Real-world applications across industries

Python Fundamentals

- Variables and data types (int, float, str, bool)
- Operators: Arithmetic and Comparison
- String operations and manipulation
- User input and type casting
- Comments and code documentation

Homework & Next Class

📚 Homework for Tonight:

1. Complete Tasks 3-10 from Annexure-I
2. Practice writing at least 5 different Python programs
3. Watch: Python basics tutorial (link in Jupyter Notebook)
4. Install all required software if not done in class

➡️ SOON Next Class (Day 2):

Data Structures: Lists, Tuples, Dictionaries, and Sets

Keep Practicing! Every Expert Was Once a Beginner 💪

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